AP CALCULUS AB

Unit 1 • Worksheet 1-6A • Answer key

# Worksheet 1-6A • Average Rate of Change

Answer key with worked solutions

**Directions**

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

## Question 1

NO CALCULATOR

What is the average rate of change of $f(x) = x^3 - 2x$ on $[-1, 2]$?

(A) $3$

(B) $\frac{5}{3}$

(C) $1$

(D) $-1$

**Solution**

**Answer: (C).**

## Question 2

NO CALCULATOR

For $f(x) = x^2 + x$, which is $\dfrac{f(2+h) - f(2)}{h}$ for $h \neq 0$?

(A) $5$

(B) $5 + h$

(C) $4 + h$

(D) $h^2 + 5h$

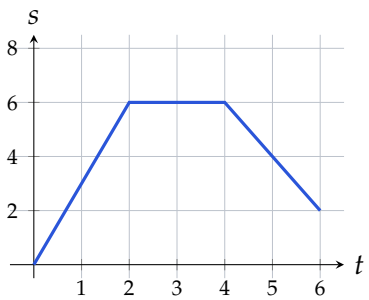
**Solution**

**Answer: (B).** $\frac{(2+h)^2+(2+h)-6}{h} = \frac{5h+h^2}{h}$.

## Question 3

NO CALCULATOR

The position (m) of a particle is graphed. What is its average velocity on $[2, 6]$?



(A) $-1$ m/s
(B) $-2$ m/s
(C) $1$ m/s
(D) $0$ m/s

**Solution**
**Answer: (A).** $\frac{2-6}{4} = -1$.

### Question 4

NO CALCULATOR

$V(t)$ is a volume in liters, $t$ in minutes. Units of $\frac{V(8)-V(2)}{6}$?

(A) liters
(B) minutes per liter
(C) liters per minute
(D) liters per minute per minute

**Solution**
**Answer: (C).**

### Question 5 (Free response)

NO CALCULATOR

| $t$ (hours) | 0 | 4 | 8 | 12 | 16 |
|---|---|---|---|---|---|
| $T(t)$ (°F) | 52 | 48 | 57 | 71 | 66 |

(a) Find and interpret the average rate of change of $T$ on $[4, 12]$.
(b) On which interval of length 4 did $T$ increase fastest on average?
(c) For $s(t) = t^3$, simplify $\frac{s(1+h)-s(1)}{h}$ and evaluate it for $h = 0.1$.

**Solution**
(a) $\frac{23}{8} = 2.875$ °F per hour, on average, from 4 a.m. to noon. (b) $[8, 12]$ (3.5). (c) $3 + 3h + h^2$; 3.31.